A SEPARATED FINITE TYPE SCHEME OF DIMENSION AT MOST ONE ADMITS AN H-PROJECTIVE COMPLETION WHOSE BOUNDARY IS A FINITE SET OF CLOSED POINTS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Scheme theoretic image and the closure of a quasi-compact immersion. Stacks Project, Morphisms, Definition morphisms-definition-scheme-theoretic-image (tag 01R6) and Lemma morphisms-lemma-quasi-compact-immersion (tag 01R8). Produces the completion with the stated density and H-projectivity properties.
- Quasi-projectivity of a separated finite type scheme of dimension at most one. Stacks Project, Varieties, Lemma lemma-dim-1-quasi-projective (tag 0A2S). Provides the initial locally closed embedding into projective space.

Source reference. Original lemma 0A25: sources/stacks-chow/tags.txt, source lines 12693–12693. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01R7: sources/stacks-chow/tags.txt, source lines 2292–2292. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01RG: sources/stacks-chow/tags.txt, source lines 2301–2301. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0A21: sources/stacks-chow/tags.txt, source lines 12689–12689. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a separated scheme of finite type over k. If $\dim(X) \leq 1$, then there exists an open immersion $j : X \to \overline{X}$ with the following properties*

(1) *$\overline{X}$ is H-projective over k, i.e., $\overline{X}$ is a closed subscheme of $\mathbf{P}^d_k$ for some d,*
(2) *$j(X) \subset \overline{X}$ is dense and scheme theoretically dense,*
(3) *$\overline{X} \setminus X = \{x_1, \ldots, x_n\}$ for some closed points $x_i \in \overline{X}$.*

Proof. By Lemma 0A25 we may assume X is a locally closed subscheme of $\mathbf{P}^d_k$ for some d. Let $\overline{X} \subset \mathbf{P}^d_k$ be the scheme theoretic image of $X \to \mathbf{P}^d_k$, see Morphisms, Definition 01R7. The description in Morphisms, Lemma 01RG gives properties (1)

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and (2). Then $\dim(X) = 1 \Rightarrow \dim(\overline{X}) = 1$ for example by looking at generic points, see Lemma 0A21. As $\overline{X}$ is Noetherian, it then follows that $\overline{X} \setminus X = \{x_1, \ldots, x_n\}$ is a finite set of closed points. $\square$